

Beginners Guide to Multirotors

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Introduction:

My name is Sean Foster and if you are reading this, then you have helped fund the kickstarter campaign for the ATLAS multirotor. I would like to thank you first and foremost for your support! Without your support, this project would not have been as successful as it is today. Our team would like to thank you from the bottom of our hearts. I hope that this guide will teach you everything you need to know about multirotors. This guide was written from the stance of someone who would like to build one for the first time, so it should have everything you need to know to build your own.

Let's get started:

I am sure you have seen some sweet quadrotor videos on youtube, and if you haven't, go check them out! In all seriousness, the first question you have to ask is, "What would I like to do with my quadrotor?" You don't have to decide now, but if you plan on building one, you should have a general idea of what you want to do. There are several different routes that you can go to build a pretty sweet quad. You can build a multirotor for professional filming, first person view racing, or to do tricks. But, as with most things in life, you have to pay to play. The budget you set aside will determine what kind of functionality and performance your new quad will have. Even though prices can vary wildly, all multirotors have the same basic parts.

Multirotor basic parts:

- Frame
- Transmitter and Receiver
- Flight Controller
- Speed Controls (ESC's)
- Motors
- Propeller mounting hardware
- Propellers
- Battery
- Servo (If building a Tricopter)
- Power Distribution Board (If not already built into the frame)

The above parts are common for every multirotor. If you pick a Hexacopter to build, that means you will have to buy 6 of almost everything. If you were to build a Tricopter instead, you would only have to purchase 3 motors, 3 speed controls, and one servo (a device that moves a horn or rod using controller inputs). Depending on what your budget is, and what you would like to do, you can now start to get an idea of what you would like to make. Let's start with some simple definitions.

- Tricopter- A Tricopter uses a central frame that has three main arms that make the shape of a Y. The front two arms provide the lift, and the rear arm has a servo that can tilt the back motor to the left and right giving the ability to turn.
- Quadcopter- A Quadcopter uses four fixed main arms that extend from a central body. With a motor at the end of each arm, all four motors generate lift. Varying the speed at which the blades spin, you get your maneuverability. A quad rotor can be configured in several basic shapes. They can be flown in an X configuration, + configuration, and H configuration. See pictures below for examples.
- Hexacopter- A Hexacopter uses 6 motors to provide a lot of lift. There are two basic configurations for a Hexacopter. The first is the traditional six arms in an asterisk shape (*), with an arm in six different directions. The second is the Y6 approach. This is basically a Tricopter with two motors on each arm. The propeller combination makes it so each motor works in unison to provide more lift.
- Octocopter- An Octocopter uses 8 motors to provide lift. Just like the Hexacopter, the Octocopter has two configurations. It can have 8 arms that jut out from the main frame, each with their own motor or, you can make an Octoquad. An Octoquad places an extra motor underneath the motors of a Quadcopter. Like the Y6, the X8 uses special propeller combinations to create lots of lift in that configuration.

Below are a few pictures from the 3DRobotics website. Disregard the propeller directions if you are not using an Ardupilot flight controller board, as different flight controllers may use different propeller combinations. This will show you what the general motor layout and frame for each type of multirotor looks like.

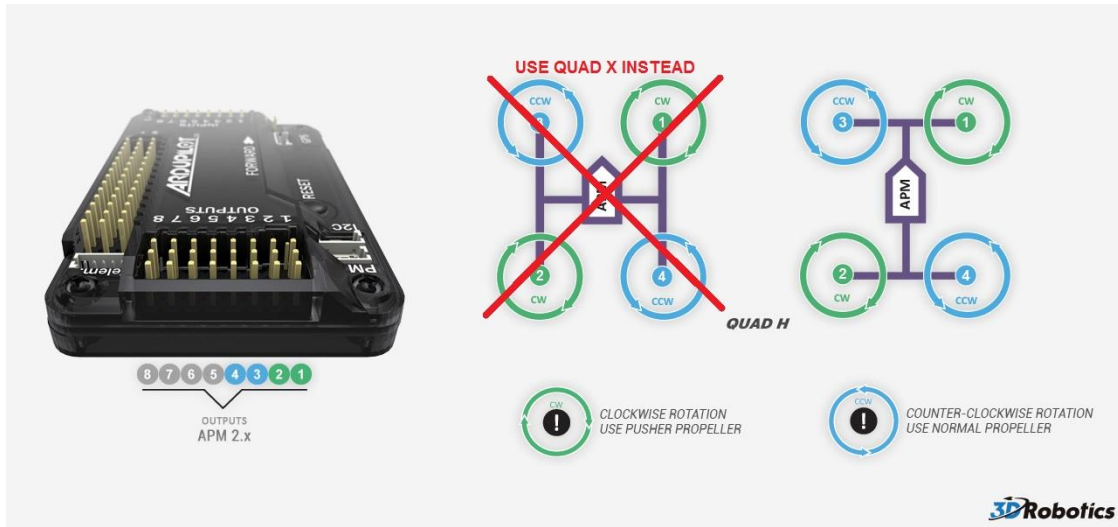
TRICOPTER



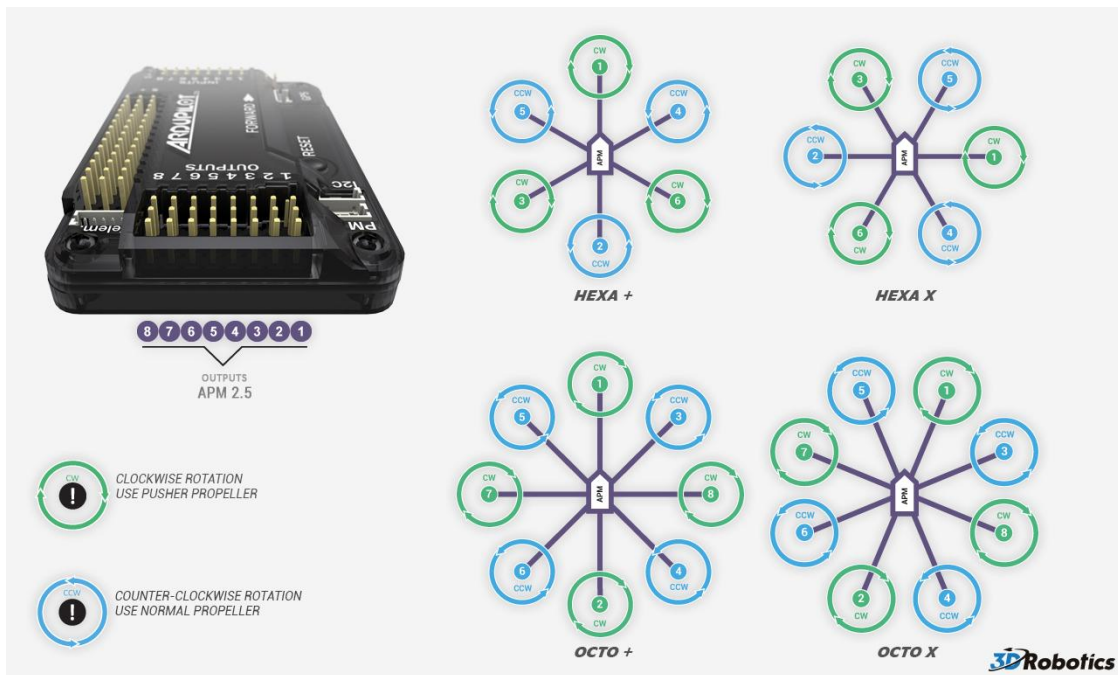
QUADCOPTER + AND X



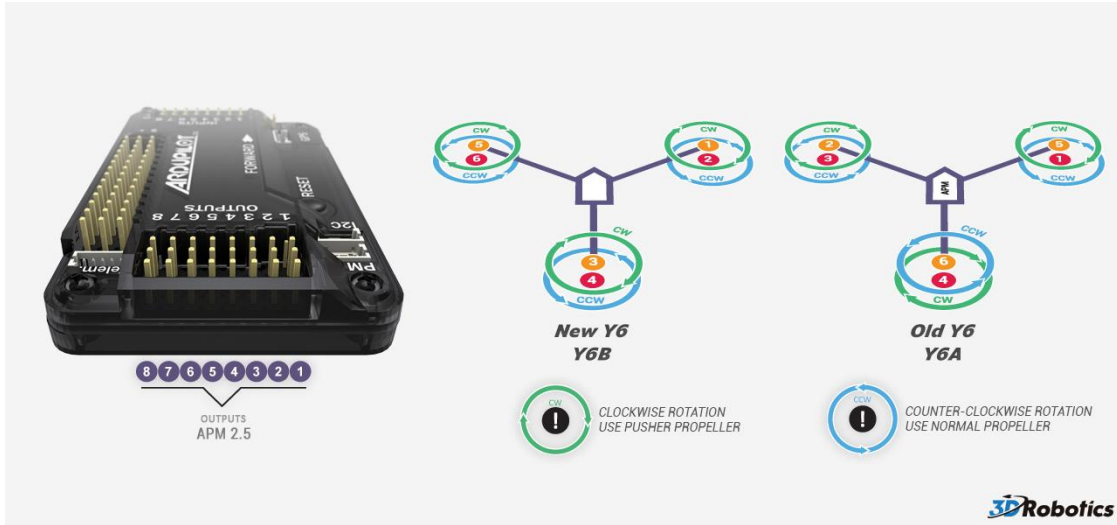
H-QUADCOPTER



HEXACOPTER & OCTOCOPTER



Y6 HEXACOPTER



X8 OCTOCOPTER



Different sizes of Multirotors

Now that you know the different types of Multirotors, it's time to pick a size. Multirotor sizes are measured by the length from the outside of one arm to the outside of the opposing arm. These sizes are normally measured in millimeters (mm). If you see a quadrotor for sale that is a 450, that means it's 450mm across. Knowing these numbers can help you determine how big you want to go, and if there is enough room to fit all your gear on it.

- A micro sized multirotor can have a "wingspan" of around 250mm.
- A small sized multirotor can have a "wingspan" of around 300-400mm.
- A medium sized multirotor is usually around the 450mm mark.
- A large sized multirotor is normally 550+.

The terms micro, small, medium, and large are used subjectively. What is large to one person may be medium to another. So I am going to give context to what these sizes and numbers mean, by application.

- A micro sized multirotor should be flown indoors and not outside. It is small enough that there is not a great danger of breaking things while flying inside, but is also too small to handle windy outside conditions.
- A small sized multirotor can be flown inside, but may cause damage to property indoors in a crash. They can also be flown outdoors and should handle wind fairly well.
- A medium sized multirotor should not be flown indoors. It can cause severe damage to person and property in an accident. It should be able to handle outside conditions without any problems, and may also carry other items.
- A large sized multirotor cannot be flown indoors. In normal situations, there is not enough room indoors to fly. It performs best outside where it is able to use its' size to do things that the medium sized may have problems doing. Even when outdoors, extreme caution must be taken to avoid accidents with copter.

I hope that gave you a little context as to about what size copter you may want to build. This brings us to the next section of this tutorial, choosing your frame!

Choosing the frame that's right for you:

There are many different configurations of multirotors out there. Hopefully by now you have an idea on the size and shape of the multirotor you would like to build. If you have already picked which frame you would like to start with, then you should stop and consider the pros and cons of each frame and configuration. Let's start with the table below.

Frame Configuration	Pros	Cons
Tricopter in Y config.	• No boom in shot if you want to film. • Potentially less weight for longer flight times • Awesome yaw control • Initial costs are low	• Turning relies on a servo which can draw a lot of power • Not as much lift as other copters • Tail servos can be expensive
Quadcopter in + config.	• Startup cost are relatively low • Can be great for maintaining orientation. • Can mount to the forward and tail booms. • Great pitch and roll maneuverability	• If you want to film, you have a boom in the shot. • Can be jerky to control • Less responsive yaw control
Quadcopter in X config.	• Startup cost are relatively low • Wide angle filming capable • Smooth flight by keeping all motors on at once	• Orientation can be harder to keep track of • Less agile than + config. • No motor redundancy • Less responsive yaw control
Quadcopter in H config.	• Nimble and responsive to rolls. • No boom in camera shot	• May take more tuning to make super stable than + or x
	Pros	Cons

Hexacopter * (6) asterisk config.	• Lots of lifting power • Frames typically are very spacious • Motor redundancy to protect against failures • Stable flight characteristics	• Battery life is poor • Startup costs are higher • Beams in shot at many angles
Hexacopter Y6 config.	• Lots of power • Condensed electronics • No beams in shot • Motor redundancy to protect against failures	• Requires a powerful servo to turn the tail boom • Battery life • Startup cost are high
Octocopter * (8) asterisk config.	• Can be used to carry very large payloads including camera gear. • Extra motor redundancies • Very stable flight	• Potential vibration problems from all motors and props • Very high startup costs • Low battery life • Time consuming setup
Octocopter X8 (dual quad) config.	• Can carry large loads • Extra motor redundancies • Only need to dampen vibrations with 4 arms instead of 8.	• Less smooth as compared to the asterisk design. • Large startup costs • Low battery life • Time consuming setup

There are a lot more frame configurations other than the ones listed, but the above are the most common. After looking at the pros and cons, and thinking about what you would like your multirotor to do you should go ahead and look at frames from online vendors. I order most of my parts from www.hobbyking.com. As a disclaimer, most of the parts that I use for all of my hobbies come from this site, so I am rather biased towards it. They have good products and great prices, just be sure to read the reviews first. There are many websites to order multicopter parts from, but each have their pros and cons. These are just a few to get started.

- Hobbyking.com
- RCtimer.com
- Foxtechfpv.com
- Flitetest.com

- FPVlab.com
- Hobbywireless.com
- Myrcmart.com
- Readymaderc.com
- Hobby-wing.com
- Store.3drobotics.com
- DIYquadcopters.com
- Mulitrotorsuperstore.com

This is a non-exhaustive list of just some vendors that I have used, or are very common in the hobby. Most of these vendors are located in china, so your parts may take longer to reach you if you are in the USA. Websites like hobbyking.com and store.3drobotics.com have warehouses in the USA, if not already located here. Even though it may cost more for parts, ebay.com is also a huge supplier of RC multicopter parts.

Components

This section will cover exactly what components are needed to make a multirotor. By now you should have an idea of what you are going to do with your copter. You first start out by choosing a frame. The frame will be determined by what you are going to do with the copter and will give you an idea of the components needed. It will also give you an idea of how much the entire system is going to weigh. Weight plays a massive part in building a multirotor so picking the proper frame, for what you want to do, is very important. Picking a frame that is too large will decrease your flight time, but picking one too small may make your components cramped. Don't be afraid to look at other builds online to see what a frame is capable of, and what components others are using. Once you have found a frame that you like it is time to move to the next component, the motors.

Motor Selection

Motor selection is the most important part of building a multirotor. Finding a frame that will suit your needs is important, but the motors determine the rest of your components. There are what you would call "Name plate ratings" for all brushless motors. These are the size, speed, voltage rating, current rating, and winding configuration of the motor. When you search for a brushless motor, there are normally three numbers that you will see, for example: "2847 – 1000kV or 28-47 1000kV." When we break that apart we get:

28= The Diameter of the motor in mm (millimeters).

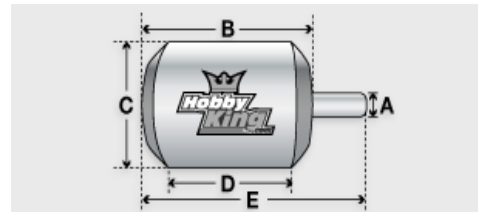
47= The Length of the motor in mm.

1000kv= Number of revolutions the motor will turn if you apply 1 volt to it.

In the above example, the numbers for the motor *actually* refer to the stator (the inside part with all the copper wire) and not the physical outside motor we interact with.

DISCLAIMER - Since the invention of the brushless motor there has not been a standardized way of measuring and classifying them, so sometimes the ratings given are not the actual size of the motor. These ratings are a good way of determining size, and comparing them to other motors.

Now that you know what the initial ratings mean, lets dive in a little more to figure out exactly what we are looking for when we build a mulitrotor. This is a commonly used multirotor motor pulled strait from the HobbyKing website with all the associated data.



Turnigy Multistar 2213-980Kv 14Pole Multi-Rotor Outrunner

Spec:

KV(RPM/V): **980**

Lipo cells: **2-3s**

Max Watts(W): **165**

Max. current(10s): **15 amps**

No Load Current: **0.5 amp**

Internal Resistance: **0.145 ohm**

Number of Poles: **14**

Shaft Size: **5mm**

Dimensions(Dia.xL): **28 x 29 mm**

Bolt Holes Spacing: **16 & 19mm**

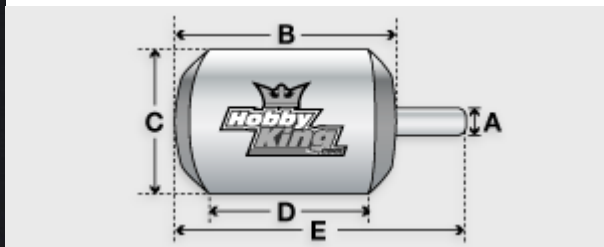
Bolt thread: **M3**

Connector: **3.5mm Bullet Type Connector**

Weight: **58g**

Kv (rpm/v)	980
Weight (g)	58
Max Current (A)	15
Resistance (mh)	145
Max Voltage (V)	11
Power(W)	165
Shaft A (mm)	0
Length B (mm)	31
Diameter C (mm)	28
Can Length D (mm)	16
Total Length E (mm)	31

When we look at the motor above we see a lot of very useful information. The first would be the size, 2213 means the stator is 22mm in diameter and 13mm high. If we look to the table on the right we can see that the diameter is actually 28mm and can length is 16mm. The next most important piece of information is the kV rating. When you have a lighter multirotor you usually want a higher kV rating with a smaller propeller. Conversely, when you have a large and heavy multirotor, you will want a lower kV rating and a larger propeller. There are no hard and fast rules stating what motor you have to use, but there are logical guidelines to follow. Most motors should come with propeller thrust figures. If we take a look at the NTM Prop Drive 28-36 motor on Hobbyking's website, they list the thrust figures for various propeller types.



Specs:

Model: **NTM Prop Drive Series 2836 1000kv**

Kv: **1000rpm/v**

Poles: **3**

Motor Wind: **14T**

Max current: **30A**

Max Power: **400W**

Shaft: **4mm**

Weight: **89g**

ESC: **30~40A**

Cell count: **3s~6s Lipoly**

Bolt holes: **16mm & 19mm**

Bolt thread: **M3**

Connection: **3.5mm Bullet-connector**

Kv (rpm/v)	1000
Weight (g)	89
Max Current (A)	30
Resistance (mh)	0
Max Voltage (V)	23
Power(W)	400
Shaft A (mm)	4
Length B (mm)	36
Diameter C (mm)	28
Can Length D (mm)	25
Total Length E (mm)	56

Prop Test Data:

Size Voltage Wattage Current Thrust

11x7 - 11.1V / 220W / 20A / 976g thrust

11x7 - 14.8V / 412W / 27.7A / 1.485kg thrust

12x6 - 11.1V / 215W / 20A / 1.08kg thrust

12x6 - 14.8V / 430W / 29A / 1.42kg thrust

13x4 - 11.1V / 223W / 20A / 1.06kg thrust

13x4 - 14.8V / 400W / 27A / 1.48kg thrust

If we look at the propeller data, we can see that this motor can swing an 11x7 propeller and create 976 grams of thrust while pulling 20 amps. If you wanted to use this motor to build a multirotor, you would want your multicopter to have a total weight of about half of your total thrust. For a quad rotor using this motor, you would shoot for an AUW (all up weight) of about 2kg or about 4 pounds. For best performance and flight time, you normally want a 2:1 power ratio. Not all motors come with propeller data, so it can be hard to discern whether a motor will work for your copter. You can use an online calculator (<http://www.ecalc.ch/>) to see if your components will work together. Ecalc does require a small fee to use, but it is fairly cheap (at about 2-3 USD a month) and will save you lots of money in the long run by helping you buy the optimal components. You should also use Ecalc to find the optimal propeller to buy for each motor.

Speed Controllers

Speed controllers, commonly abbreviated to ESC (electronic speed controller) are used to power the motors on a multirotor. There are a ton of different types and sizes of speed controls on the market. The first step to finding the proper speed controller is finding the motor you are going to use. In this guide we are only discussing brushless speed controls and motors. Brushed motors are really outdated and give poor performance so be sure that you buy brushless equipment when building a multirotor. If you look at the above motor example you can see that if you install an 11x7 propeller, you will pull about 20A at full throttle. You would then want to buy a speed controller that is rated for about 1.5x the max current for the motor at the recommended voltage. If you wanted to use the above motor you should use a speed controller rated for about 30 amps to be safe. This will also give you room to upgrade to a larger propeller and will keep the ESC running cooler. It never hurts to go larger than expected, except maybe the wallet.

I am not going to get into the programming portion of speed controls, but it is possible to put custom software on some speed controllers. The Afro series of speed controls from hobbyking come pre-flashed with custom SimonK firmware which makes them more efficient and responsive. That being said, there is not much difference between ESC's for planes and multirotors. They are interchangeable, but normally, the controllers meant for multicopters provide increased performance and efficiency. There is an example of an Afro series ESC below.



Afro ESC 12Amp Multi-rotor Motor Speed Controller (SimonK Firmware)

Specs:

Current Draw: **12A Continuous**

Voltage Range: **2-4s Lipoly**

BEC: **0.5A Linear**

Input Freq: **1KHz**

Firmware: **afro_nfet.hex**

Discharge wire/plugs: **16AWG/Male 3.5mm**

Motor wire/plugs: **16AWG/Female 3.5mm**

Weight: **20.3g (Included wire, plug, heat shrink)**

Size: **29.5 x 22.6 x 6mm**

As you can see above, the maximum current you are able to pull is about 12A. Therefore, this would not be a good speed control for our example motor. The next important component is the BEC. BEC stands for Battery Elimination Circuit. This means that when you power the speed control with the positive and negative leads to a battery, it will supply the receiver or flight controller board with power. Almost all speed controls have a BEC that outputs 5 volts. This speed control is capable of supplying a maximum of .5 amps, which is quite small in comparison to others that can supply up to 5 amps. This ESC is meant to just power the flight controller and maybe a GPS module, but nothing more. You would not want to run the tail servo on a Y copter with this speed control as it does not have enough power to run it.

There are different types of BEC's and speed controls. Some speed controls specifically say "Linear" when they list the BEC output. This means that whatever voltage (usually 12.6v from a 3 cell battery or 8.4v from a two cell battery) you are putting in is being "Stepped down" to 5 volts by burning off the difference in voltage as heat. There is another type of BEC that is "Switching." Switching mode power supplies convert incoming voltage into short pulses that "Step down" the voltage. These are more efficient because they do not burn off extra voltage as heat, but rather turn on and off very fast to provide exactly the voltage you need. There is one more type of speed controller denoted as "OPTO."

OPTO speed controllers do not have onboard voltage regulators to supply voltage. Normally, you would have one speed control that has a “Switching” or “Linear” regulator to provide power to everything, and then have the rest of the speed controls as “OPTO.” It is normally not a good idea to have more than one speed control to power the same components. Linear BEC’s *can* run in parallel, although it is not recommended. Switching types *cannot* be put in parallel. You can modify a normal speed control to OPTO by cutting the middle red wire on the servo lead coming from it. Insulate the newly cut wire with electrical tape or heat shrink and you now have an OPTO ESC.

Flight Controllers

This section is devoted to the “Brains” of the multirotor. A flight controller takes in the stick movements of your transmitter, and then sends commands to the speed controls to make the motors move properly. There are a lot of flight controllers on the market right now so it is nearly impossible for me to cover every detail about every flight controller so I will just discuss the similarities and commonalities of flight controllers.

Almost every flight controller has a gyroscope and an accelerometer. These two devices control the orientation of the copter. These two devices are the bare minimum for having a flight controller board. After these two components are installed, any other sensors start to increase features and functionality. A compass and magnetometer help aid with direction and heading. A barometer and sonar will give you altitude readings and height controlling features. A GPS will give you positional data as well as heading, speed, and altitude readings. Adding a GPS greatly increases the potential of a flight controller board, as most are able to fly autonomously once equipped.

Most flight controllers have programs that are used to program different rates and settings in. For example, the Ardupilot and Pixhawk all use the Mission Planner software. There are certain boards like the Hobbyking KK2.1.5 that have a screen that you can adjust and setup the board with. Different copters require different amounts of inputs, so make sure that if you are running an Octocopter, that your board has at least 8 outpin pins.

Each flight controller board must be configured and “tuned” for the copter that it is on. Each controller board has what are called “Gains.” These gains control how fast and slow your copter reacts to your inputs and outside forces like wind. I am not going to get into gains a whole lot in this tutorial because they are very specific to each individual copter and controller board. There are many resources online on how to tune the copter to the flight controller you are using. If you purchased a ready-to-fly (also known as an RTF) copter, these gains may be set and unchangeable.

As a rule of thumb, it is always better to be able adjust the gains, than have them permanent. Also, as you add weight or change propellers you may need to adjust the gains to make it more agile or even flyable.

When considering building a multirotor, choose the features that are the most important to you, and then do a search on some of the sellers online. User friendliness, online support, manufacturer support, features and compatibility are some of the major areas that should be considered when buying a flight controller.

Batteries

Now we come to the last important component, the battery. Batteries range in size, shape, voltages, cell counts and chemistry. In order to find a battery that will fit the multirotor we want to build, we have to look at the speed controller and motor ratings. The voltage of our battery has to match that of our motor and speed controller. The batteries that are the most common in the multirotor hobby are lithium polymer batteries. These are most often shortened to LiPo. LiPo batteries consist of small packets filled with different chemicals that store charge. These packets are known as cells. Cells can be added in series or parallel to change the characteristics of the battery. Added cells in series (S) increase voltage and added cells in parallel (P) increase capacity and do not change voltage. LiPo batteries also have other properties such as; total capacity, discharge rate, burst rate, and charge rates. But, before we get into those properties, we are going to talk a little about cells and cell configurations. Every LiPo cell has a nominal charge of 3.7 volts. This is when the cell is discharged and no longer capable of providing significant power. When a LiPo cell is fully charged it should read 4.2 volts. If you combine cells in series you get different voltages. Please refer to the table below.

Number of Cells in Series	Nominal Voltage	Fully Charged Voltage
1S	3.7	4.2
2S	7.4	8.4
3S	11.1	12.6
4S	14.8	16.8
5S	18.5	21
6S	22.2	25.2

This table can scale into infinity if you wanted it to, but it is very rare to see a multicopter with anything larger than a 4 cell battery on board. You should also take special consideration when choosing a battery to make sure all of your components are okay with taking the voltage of your battery. In order to discuss the other features of a LiPo battery, let's take a look at a 4 cell battery from Hobbykings website.



Turnigy 2200mAh 4S1P 20C Lipoly Pack

Capacity: **2200mAh**
 Constant discharge: **20C**
 Burst rate: **30C (15sec)**
 Configuration : **4S 14.8v**
 Pack size: **105x34x33**
 Weight : **247g**

Capacity(mAh)	2200
Config(s)	4
Discharge(c)	20
Weight(g)	247
Max Charge Rate (C)	4
Length-A(mm)	105
Height-B(mm)	33
Width-C(mm)	34

If we look at the pack above, we can see that it has quite a few numbers and attributes. Let us start first with the pack configuration. This pack is in a 4S1P configuration, which means, we have 4 cells that have been put in series, and 1 cell that has been put in parallel with another cell. It has a total capacity of 2200 milliamp hours, and has a constant discharge rate of 20C. These two numbers are very important to multirotors. The capacity (mAh) is how long we can fly and the C rating is how much power it can constantly give. The burst rate is how much the battery can give for 15 seconds under heavy load. We can use these two numbers to give us the amount of Amps this battery can provide constantly and under burst conditions.

1. Take the capacity in milliamps (in this case 2200) and divide by 1000.

* This result will give us the capacity in amps. If you know the amps, skip step 1.

2. Now, multiply the constant C rating by the amp rating we just calculated.

- Constant Amp Discharge = (Constant C rating) X (Amp Rating)
- Constant Amp Discharge = 20 X 2.2A = 44A

Now that we know what the constant Amp discharge of the battery is, we can discern whether this battery will suit our needs. Do the combined current outputs of your

motors exceed 44 amps? Are your speed controls rated to handle 11 Amps constantly? These are just a few questions to ask yourself when buying a battery. The burst current is important to make sure that if we do “punch it” we will have enough discharge to power our motors. It is calculated the exact same way as the constant discharge but using the burst C rating. There is also a charge C rate that is associated with batteries. Some batteries can be charged at 8-10C but it wears on the internal chemistry so it is normally best if you charge at 1 to 2C.

WARNING!!! *ALWAYS USE A LIPO CHARGER FOR CHARGING LIPO BATTERIES*

WARNING!!! *NEVER OVER CHARGE OR DISCHARGE YOUR LIPO BATTERIES*

WARNING!!! *MISUSE OF LIPO BATTERIES CAN RESULT IN VIOLENT FIRES*

Sorry, I had to get that out of my system. LiPo batteries can be extremely dangerous if not handle properly. If your LiPo packs start to get puffy or round then discard of them in a safe and responsible manner. There are recycling centers for LiPo batteries so keep that in mind when discarding them.

Conclusion

I hope this guide has taught you everything that you need to know about multirotors. You should have all of the knowledge needed to build your own! I am sure that there are some small nuances that I have not covered, but that is the fun of having a hobby; learning as you go! Also, I would like to thank you again for supporting the ATLAS project on kickstarter and I really appreciate all of you for making my project a reality. I hope that everyone who reads this will get into the hobby, and I hope to fly with you all one day.

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